ModelIt!

Student Activity Pack

Your Phone's Dirty Secret

The Hidden Environmental Cost of the Device in Your Pocket

Grade 8 | NGSS MS-ESS3-3, MS-ESS3-4 | Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Where do you think the materials in your phone come from?

2 What happens to a phone when you throw it away?

3 How many phones do you think are sold worldwide each year?

4 Draw the journey of a smartphone from raw materials to your pocket to the trash.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Consumer Demand | Mining Intensity | Environmental Damage | E-Waste Accumulation

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Current Trends

Normal demand and mining levels

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Maximum Consumption

Lock Consumer Demand to maximum

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Reduced Demand

Cut Consumer Demand by 50%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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> Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Recycling Rate — Percentage of old electronics properly recycled for materials

Carbon Emissions — CO2 released during mining, manufacturing, and shipping

Water Pollution — Toxic chemicals from mining that contaminate local water sources

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

What is the true environmental cost of the phone in your pocket? Use evidence from your simulation to support your answers.

1. **How does your personal technology use contribute to environmental damage?**

2. **Why do you think only 20% of e-waste is properly recycled?**

3. **Is it possible to have modern technology WITHOUT environmental destruction? How?**

4. **Who bears the greatest cost of smartphone production — consumers or mining communities?**

5. **What would you be willing to change about how you use technology?**

8 STEM Challenge: Design a Sustainable Phone Program

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YOUR ENGINEERING MISSION

Design a school or community program that reduces the environmental impact of smartphone consumption.

THE SCENARIO

Your school district wants to reduce its technology footprint. Design a program that balances student access to technology with environmental responsibility.

Engineering Design Thinking

1. HOW WILL YOUR PROGRAM REDUCE NEW DEVICE PURCHASES?
2. WHAT'S YOUR PLAN FOR RESPONSIBLE E-WASTE DISPOSAL?
3. HOW WILL YOU EDUCATE THE COMMUNITY ABOUT ELECTRONIC WASTE?
4. HOW WILL YOU MEASURE THE ENVIRONMENTAL IMPACT OF YOUR PROGRAM?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Environmental Engineers and Sustainability Consultants help companies reduce their environmental footprint and design greener products. They earn $70,000-$120,000/year.